Engineering Mathematics

Mid-term Review

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Chapter 1 Complex Number

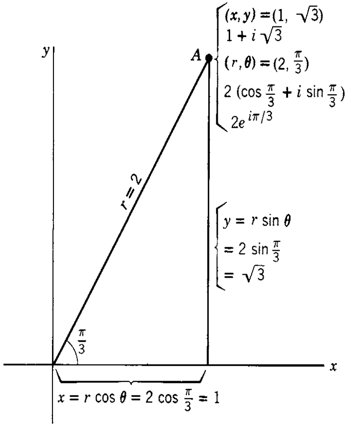
Complex representation

- Real-imaginary part representation z= a + bi
- Polar representation z = r $e^{i\theta}$
- Geometric representation
- Modulus, angle (argument)

$$i^2 = -1$$

$$x + iy = r(\cos\theta + i\,\sin\theta) = re^{i\theta}.$$

$$\begin{aligned} &\operatorname{Re} z = x, && |z| = \operatorname{mod} z = r = \sqrt{x^2+y^2}, \\ &\operatorname{Im} z = y\,(\textbf{not}\ iy), && \text{angle of } z = \theta. \end{aligned}$$



Operation

- Addition, substraction (opposite couterpart), multiplication, division
- Complex conjugate z^* = a − bi, z^* = r $e^{-i\theta}$

$$z \cdot z^* = x^2 + y^2 \geq 0$$

$$\frac{z_1}{z_2} = \frac{x_1 + \mathrm{i}y_1}{x_2 + \mathrm{i}y_2} = \frac{(x_1 + \mathrm{i}y_1)\boxed{(x_2 - \mathrm{i}y_2)}}{(x_2 + \mathrm{i}y_2)\boxed{(x_2 - \mathrm{i}y_2)}}$$

$$= \frac{(x_1x_2 + y_1y_2) + \mathrm{i}(x_2y_1 - x_1y_2)}{x_2^2 + y_2^2}$$

$$\frac{z_1}{z_2} = \frac{r_1e^{\mathrm{i}\theta_1}}{r_2e^{\mathrm{i}\theta_2}} = \frac{r_1}{r_2}e^{\mathrm{i}(\theta_1 - \theta_2)} \qquad (z_2 \neq 0)$$

Graphically, it means the final modulus = $\frac{r_1}{r_2}$ (模相除)

the final argument = $\theta_1 - \theta_2$ (辅角相减)

As usual we require that $z_2 \neq 0$.

Conversion between real-imaginary part representation and polar representation

Problem 1. Transform the following into the form of $x + iy$, and find their absolute values.

(a) (10 marks) $\frac{3+2i}{1-i}$

(b) (10 marks) $\cos(i \ln i)$

$$\frac{1}{0.5(\cos 40° + i \sin 40°)}$$

Complex equations

Let's recall we have learned before:

$z_1 = x_1 + \mathrm{i}y_1$ and $z_2 = x_2 + \mathrm{i}y_2$

$z_1 = z_2$, if and only if $\underline{x_1 = x_2}$ and $\underline{y_1 = y_2}$ Note: there are actually two equations

Using the above relations, we can work with complex equations.

一个复等式 相当于两个实等式

4. Solve for all possible values of the real numbers x and y in the following equations.

   (a) $(2x - 3y - 5) + i(x + 2y + 1) = 0$
   (b) $\frac{x+iy}{x-iy} = -i$
   (c) $|1 - (x + iy)| = x + iy$

Euler formula, De Moivre's Theorem

- Let's do the Taylor expansion for triangular functions around $\theta = 0$.

$$\left.\begin{aligned}\cos\theta &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \ldots \\ \sin\theta &= \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \ldots\end{aligned}\right\}$$

$$\begin{aligned}\boxed{e^{\mathrm{i}\theta}} &= 1 + \mathrm{i}\theta + \frac{(\mathrm{i}\theta)^2}{2!} + \frac{(\mathrm{i}\theta)^3}{3!} + \frac{(\mathrm{i}\theta)^4}{4!} + \frac{(\mathrm{i}\theta)^5}{5!} + \ldots \\ &= 1 + \mathrm{i}\theta - \frac{\theta^2}{2!} - \mathrm{i}\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + \mathrm{i}\frac{\theta^5}{5!}\ldots \\ &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \ldots + \mathrm{i}(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!}\ldots) \\ &\boxed{= \cos\theta + \mathrm{i}\sin\theta}\end{aligned}$$

Euler's formula (欧拉公式)47

$$z^n = (re^{\mathrm{i}\theta})^n = r^n \cdot e^{\mathrm{i}n\theta}$$

Let $r = 1$

De Moivre's theorem (棣莫弗定理)

$$\boxed{(e^{\mathrm{i}\theta})^n = (\cos\theta + \mathrm{i}\sin\theta)^n = e^{\mathrm{i}n\theta} = \cos n\theta + \mathrm{i}\sin n\theta}$$

Chapter 2. Complex Function

Power function (幂函数): $f(z) = z^n \quad n \text{ is an integer}$

Exponential function (指数函数): $f(z) = 10^z \quad f(z) = e^z$

Logarithmic function (对数函数): $f(z) = \log(z) \quad f(z) = \ln(z)$

Trigonometric function (三角函数): $f(z) = \sin(z),\ \cos(z),\ \tan(z),\ \sec(z)$

Inverse trigonometric function (反三角函数):

$$f(z) = \arcsin(z),\ \arccos(z),\ \arctan(z),\ \mathrm{arcsec}(z)$$

Power function, Root function

$f(z) = z^n$ $\quad$ n is an integer

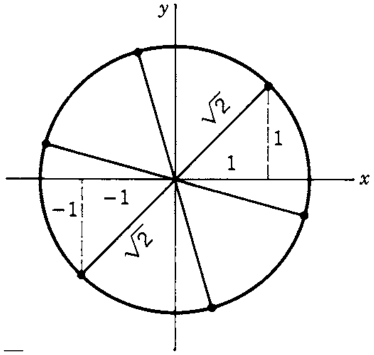
$f(z) = z^2$ $\quad$ $f(1+\mathrm{i}) = (1+\mathrm{i})^2 = 2\mathrm{i}$

Summary In each of the preceding examples, our steps in finding $\sqrt[n]{re^{i\theta}}$ were:

(a) Find the polar coordinates of the roots: Take the nth root of r and divide $\theta + 2k\pi$ by n.

(b) Make a sketch: Draw a circle of radius $\sqrt[n]{r}$, plot the root with angle θ/n, and then plot the rest of the n roots around the circle equally spaced $2\pi/n$ apart. Note that we have now essentially solved the problem. From the sketch you can see the approximate rectangular coordinates of the roots and check your answers in (c). Since this sketch is quick and easy to do, it is worthwhile even if you use a computer to do part (c).

(c) Find the $x + iy$ coordinates of the roots by one of the methods in the examples. If you are using a computer, you may want to make a computer plot of the roots which should be a perfected copy of your sketch in (b).



Exponential Function

$$e^z = \sum_0^\infty \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots .$$

$$e^z = e^{x+iy} = e^x e^{iy} = e^x(\cos y + i \sin y).$$

- A complex exponential function can be periodic (具有周期性).

$$e^z \cdot e^{2n\pi \mathrm{i}} = e^z \cdot 1 = e^z, \text{ where } n = \text{integer}$$

Trigonometric function

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}.$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i},$$

$$\sinh z = \frac{e^{z} - e^{-z}}{2}, \qquad \cosh z = \frac{e^{z} + e^{-z}}{2}.$$

- Similar to the complex exponential function, trigonometric functions are also periodic. Note: their periods are not exactly the same (mind the difference i).

$$\sin(z) = \sin(z \pm 2n\pi), \text{ and } \cos(z) = \cos(z \pm 2n\pi) \qquad \text{period} = 2n\pi$$

- Unlike real trigonometric functions, the modulus of a complex trigonometric function can be greater than 1.

$$\cos i = \frac{e^{i \cdot i} + e^{-i \cdot i}}{2} = \frac{e^{-1} + e}{2} = 1.543 \cdots.$$

- Also think about the asymptotic behavior of trigonometric function, when $z \to \infty$

$$\lim_{z = \mathrm{i}y \to +\mathrm{i}\infty} \cos(z) = \frac{e^{(-\infty)} + e^{(+\infty)}}{2}$$

Unbounded (无界的)

$$\lim_{z = x \to +\infty} \cos(z) = \frac{e^{\mathrm{i}x} + e^{-\mathrm{i}x}}{2}$$

Bounded (有界的)

Example] Solve the equation $\cos(z) = \mathrm{i}$

Solution : $\text{L.H.S} = \dfrac{e^{\mathrm{i}z} + e^{-\mathrm{i}z}}{2} = \mathrm{i}$

$e^{\mathrm{i}z} \neq 0$ multiply $e^{\mathrm{i}z}$ on both sides

recall that $\sqrt{-1} = \pm\mathrm{i}$

Then we have $e^{2\mathrm{i}z} - 2\mathrm{i}e^{\mathrm{i}z} + 1 = 0$

Let $e^{iz} = z_1$ we have $z_1^2 - 2\mathrm{i}z_1 + 1 = 0$

$$z_1 = \frac{2\mathrm{i} \pm \sqrt{-4-4}}{2} = (1 \pm \sqrt{2})\mathrm{i}$$

$=> e^{-y}e^{\mathrm{i}x} = (1 \pm \sqrt{2})\mathrm{i}$

Situation-1: $x = 2n\pi + \pi/2,\ y = -\ln(1 + \sqrt{2})$

Situation-2: $x = 2n\pi - \pi/2,\ y = -\ln(\sqrt{2} - 1)$

Logorithm Function

$$w = \ln z = \ln(re^{i\theta}) = \operatorname{Ln} r + \ln e^{i\theta} = \operatorname{Ln} r + i\theta, \tag{13.5}$$

where $\operatorname{Ln} r$ means the ordinary real logarithm to the base e of the real positive number r.

Since θ has an infinite number of values (all differing by multiples of 2π), a complex number has infinitely many logarithms, differing from each other by multiples of $2\pi i$. The *principal value* of $\ln z$ (often written as $\operatorname{Ln} z$) is the one using the principal value of θ, that is $0 \le \theta < 2\pi$. (Some references use $-\pi < \theta \le \pi$.)

Example 2. Find $\ln(1 + i)$. From Figure 5.1, for $z = 1 + i$, we find $r = \sqrt{2}$, and $\theta = \pi/4 \pm 2n\pi$. Then

$$\ln(1 + i) = \operatorname{Ln} \sqrt{2} + i\left(\frac{\pi}{4} \pm 2n\pi\right) = 0.347\cdots + i\left(\frac{\pi}{4} \pm 2n\pi\right).$$

Even a positive real number now has infinitely many logarithms, since its angle can be taken as 0, 2π, -2π, etc. Only one of these logarithms is real, namely the principal value $\operatorname{Ln} r$ using the angle $\theta = 0$.

Complex Powers

$$a^b = e^{b \ln a}.$$

Example 1. Find all values of i^{-2i}. From Figure 5.2, and equation (13.5) we find $\ln i = \operatorname{Ln} 1 + i(\pi/2 \pm 2n\pi) = i(\pi/2 \pm 2n\pi)$ since $\operatorname{Ln} 1 = 0$. Then, by equation (14.1),

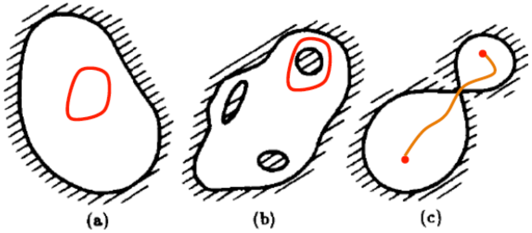
$$i^{-2i} = e^{-2i \ln i} = e^{-2i \cdot i(\pi/2 \pm 2n\pi)} = e^{\pi \pm 4n\pi} = e^{\pi}, e^{5\pi}, e^{-3\pi}, \cdots,$$

where $e^{\pi} = 23.14 \cdots$. Note the infinite set of values of i^{-2i}, all real!

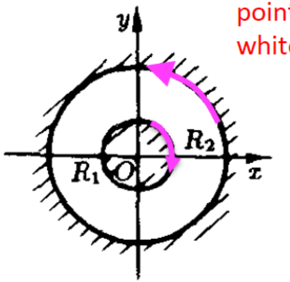
Chapter 3. Analytical Function

Set, Connectivity, Direction

- Singly connected region, multi-connected region



We consider the points in the white area.



(c) $R_1 < |z| < R_2$

What is the positive direction along R_1?

What about along R_2?

Analytical Function

Definition The derivative of $f(z)$ is defined (just as it is for a function of a real variable) by the equation

$$f'(z) = \frac{df}{dz} = \lim_{\Delta z \to 0} \frac{\Delta f}{\Delta z}, \tag{2.1}$$

where $\Delta f = f(z + \Delta z) - f(z)$ and $\Delta z = \Delta x + i\Delta y$.

Definition: A function $f(z)$ is *analytic* (or *regular* or *holomorphic* or *monogenic*) in a region* of the complex plane if it has a (unique) derivative at every point of the region. The statement "$f(z)$ is analytic *at a point* $z = a$" means that $f(z)$ has a derivative at every point inside some small circle about $z = a$.

Analytical at a point

Proof of Analytical Function

- Definition: the function has a (unique) derivative at every point in a region of the complex plane.
- CR-relation + continuity

偏导连续

> **Theorem II** (which we state without proof). If $u(x, y)$ and $v(x, y)$ and their partial derivatives with respect to x and y are continuous and satisfy the Cauchy-Riemann conditions in a region, then $f(z)$ is analytic at all points inside the region (not necessarily on the boundary).

- Morera's Theorem

Morera's theorem (莫列拉定理): If $f(z)$ is continuous in $\overline{G}$, if for any closed curve (contour) in $\overline{G}$, $\oint_C f(z)dz = 0$, then $f(z)$ is analytic in G.

Cauchy-Riemann condition

Theorem I (which we shall prove). If $f(z) = u(x,y) + iv(x,y)$ is analytic in a region, then in that region

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}. \tag{2.2}$$

These equations are called the *Cauchy-Riemann conditions.*

$$\frac{\partial u}{\partial x} \quad -\frac{\partial u}{\partial y}$$
$$\frac{\partial v}{\partial x} \quad \frac{\partial v}{\partial y}$$

- 实部和虚部的关联很强
- 解析函数的必要条件, quivalence at x-y dimension
- Polar domian

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$$

- Whether an analytical function $f(z)$ can be a function of $\bar{z}$? No.

U(x,y) and V(x,y)

- Obtain U(x,y) and V(x,y)
- The real part U(x, y) and the imaginary part V(x, y) are not independent.
  - ✓CR condition
  - ✓Find V from U for analytical function

$$dv(x,y) = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy$$

Full derivative

$$v(x,y) = \int dv(x,y) = \int -\frac{\partial u}{\partial y}dx + \frac{\partial u}{\partial x}dy$$

Property of Analytical Function

Theorem III (which we state without proof). If $f(z)$ is analytic in a region (R in Figure 2.3), then it has derivatives of all orders at points inside the region and can be expanded in a Taylor series about any point z_0 inside the region. The power series converges *inside* the circle about z_0 that extends to the nearest singular point (C in Figure 2.3).

Theorem IV. Part 1 (to be proved in Problem 44). If $f(z) = u + iv$ is analytic in a region, then u and v satisfy Laplace's equation in the region (that is, u and v are harmonic functions).

Part 2 (which we state without proof). Any function u (or v) satisfying Laplace's equation in a simply-connected region, is the real or imaginary part of an analytic function $f(z)$.

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\frac{\partial v}{\partial y} = \frac{\partial^2 v}{\partial x \partial y}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(-\frac{\partial v}{\partial x}\right) = -\frac{\partial^2 v}{\partial x \partial y}$$

Laplace's equation

$$\Delta u = \nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\Delta v = \nabla^2 v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Example of Analytical Function

- z^n
- e^z
- $\sin(z)$, $\cos(z)$
- 双曲函数

$$\sinh z = \frac{e^z - e^{-z}}{2}, \qquad \cosh z = \frac{e^z + e^{-z}}{2},$$

$$\tanh z = \frac{\sinh z}{\cosh z}, \qquad \coth z = \frac{\cosh z}{\sinh z},$$

$$\operatorname{sech} z = \frac{1}{\cosh z}, \qquad \operatorname{csch} z = \frac{1}{\sinh z}.$$

- Combinations: $f(g(z))$

Chapter 4. Contour Integration

Complex Integration

$$\int_C f(z)dz = \lim_{\max|\Delta z_k| \to 0} \sum_{k=1}^{n} f(\zeta_k)\Delta z_k$$

- 实积分:Integrated along the **real *x*-axis**.
- 复积分:Integrated over a **path in the complex plane** (known as a **contour)** 围线
- 复积分
  - 方向性？
  - 积分路径？

Complex Line Integral

- 复变积分是两个实变积分的和

$$\int_\gamma f(z)\,dz = \int_\gamma (u+iv)(dx+idy) = \int_C (udx - vdy) + \mathrm{i}\int_C (vdx + udy)$$

- 参数化表示

Line integrals are also called path or contour integrals. Given the above ingredients, we define the complex line integral by

thinking of $\gamma(t)$ as $(x(t), y(t))$

$$\int_\gamma f(z)\,dz := \int_a^b f(\gamma(t))\gamma'(t)\,dt. \tag{25a}$$

Example Compute $\int_\gamma z^{-1}\,dz$ over several contours

(i) The line from 1 to $1+i$.
(ii) The circle of radius 1 around $z=3$.
(iii) The unit circle.

Solution: For parts (i) and (ii) there is no problem using the antiderivative $\log(z)$ because these curves are contained in a simply connected region that doesn't contain the origin.

(i)

$$\int_\gamma \frac{1}{z}\,dz = \log(1+i) - \log(1) = \log(\sqrt{2}) + i\frac{\pi}{4}$$

(ii) Since the beginning and end points are the same

$$\int_\gamma \frac{1}{z}\,dz = 0$$

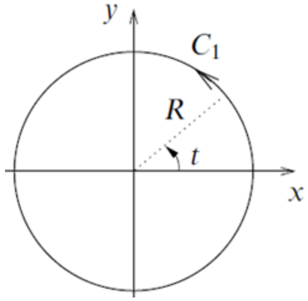
(iii) We parametrize the unit circle by $\gamma(\theta) = \mathrm{e}^{i\theta}$ with $0 \le \theta \le 2\pi$. We compute $\gamma'(\theta) = i\mathrm{e}^{i\theta}$. So the integral becomes

$$\int_\gamma \frac{1}{z}\,dz = \int_0^{2\pi} \frac{1}{\mathrm{e}^{i\theta}}\, i\mathrm{e}^{i\theta}\,dt = \int_0^{2\pi} i\,dt = 2\pi i$$

How about a circle with radius R?

闭合回路
积分一圈

$$\oint \frac{dz}{z} = \int_0^{2\pi} i d\theta = 2\pi i$$



This very important result will be used many times later, and **the following should be carefully noted**

(i) Its value, 2 πi
(ii) This value is independent of R
(iii) 不依赖于路径形状，**(**解析区域内**)**连续形变积分值不变

Cauchy's Theorem

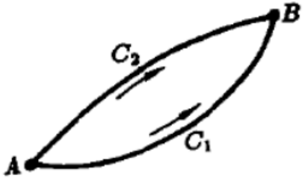
Theorem V Cauchy's theorem (see discussion below). Let C be a simple† closed curve with a continuously turning tangent except possibly at a finite number of points (that is, we allow a finite number of corners, but otherwise the curve must be "smooth"). If $f(z)$ is analytic on and inside C, then

$$\oint_{\text{around } C} f(z)\,dz = 0. \tag{3.1}$$

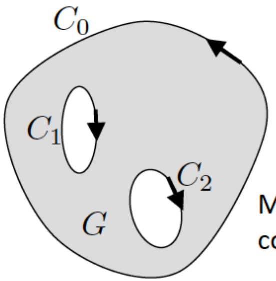
- 解析函数的积分与路径无关

If $f(z)$ is analytic in a bounded and simply connected region G, then $\int_C f(z)dz$ does not depend on the path of integration, where $C \subset G$.

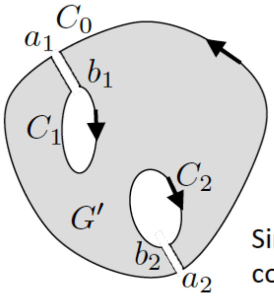
Questions: can you prove the above statement?



Extended Gauchy’s Theorem



Multi-connected



Simply connected

- Then for multi-connected region

$$\boxed{\oint_C f(z)dz = 0}$$

$$\oint_{C_0} f(z)dz + \oint_{C_1} f(z)dz + \oint_{C_2} f(z)dz = 0$$

equivalent to

$$\oint_{C_0} f(z)dz = \oint_{C_1^-} f(z)dz + \oint_{C_2^-} f(z)dz$$

Here C_i^- keeps the same direction as C_0 (i.e. clockwise or anti-clockwise)

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Example Find the value of $\oint_C z^n dz$, where *n* is an integer, *C* is a simply closed curve in $\mathbb{C}$.

1. If *n* is non-negative, z^n is analytic, then $\oint_C z^n dz = 0$.

2. If *n* is negative, and if the contour does not enclose $z = 0$, then z^n is analytic inside the region bounded by *C*, and again we have $\oint_C z^n dz = 0$.

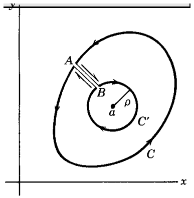
3. If *n* is negative, and if the contour encloses $z = 0$. We can draw a simple circle around $z = 0$, and

$$\oint_C z^n dz = \oint_{|z|=\varepsilon} z^n dz = \int_0^{2\pi} \varepsilon^{n+1} e^{\mathrm{i}(n+1)\theta} \mathrm{i} d\theta = \begin{cases} 2\pi \mathrm{i}, & n = -1; \\ 0, & n = -2, -3, -4, \ldots; \end{cases}$$

Cauchy Integral Formula

Theorem VI Cauchy's integral formula (which we shall prove). If $f(z)$ is analytic on and inside a simple closed curve C, the value of $f(z)$ at a point $z = a$ inside C is given by the following contour integral along C:

$$f(a) = \frac{1}{2\pi i}\oint \boxed{\frac{f(z)}{z-a}}dz.$$



1. This formula is saying that the value of an analytic function at any point inside a closed contour is uniquely determined by its values on the contour and that the specific expression above can give the value at the interior
2. It says that knowing the values of f on the boundary curve C means we everything about f inside C!!

边界(的性质)决定内部

$$f(z) = \frac{1}{2\pi i}\oint_C \frac{f(w)}{w-z}\,dw, \qquad z \text{ inside } C.$$

Higher-Order Derivatives of Analytical Function

Then $F(z) = \int_a^b f(t,z)dt$ is analytic in G, and

$$F'(z) = \int_a^b \frac{\partial f(t,z)}{\partial z} dt, \ \ z \in G$$

- Based on the Cauchy integral formula, we can deduce a very important conclusion:

If $f(z)$ is analytic in a bounded region $\overline{G}$, then any higher-order derivative of $f(z)$ exists, which is given by

$$f^{(n)}(z) = \frac{n!}{2\pi \mathrm{i}} \oint_C \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta$$

Exercise

$$\oint_C \frac{\sin 2z \, dz}{(6z - \pi)^3} \text{ where } C \text{ is the circle } |z| = 3.$$

Chapter 5. Taylor Series, Laurent series

Series Convergence

Preliminary test. If the terms of an infinite series do *not* tend to zero (that is, if $\lim_{n\to\infty} a_n \neq 0$), the series diverges. If $\lim_{n\to\infty} a_n = 0$, we must test further.

For series of positive terms

- **Comparison method**

  If $\exists N \in \mathbb{N}$, $\forall n > N$, the condition $|u_n| < v_n$ is satisfied. If $\sum_{n=0}^{\infty} v_n$ are convergent, then $\sum_{n=0}^{\infty} |u_n|$ are convergent.

- **Ratio method**

  If there exists a constant ρ (un-correlated with n), and $|u_{n+1}/u_n| < \rho < 1$, then $\sum_{n=0}^{\infty} u_n$ are absolutely convergent.

- **d'Alembert method (criterion)**

  $\overline{\lim}_{n\to\infty} |u_{n+1}/u_n| < 1 \longrightarrow \sum_{n=0}^{\infty} u_n \longrightarrow$ absolutely convergent

  $\underline{\lim}_{n\to\infty} |u_{n+1}/u_n| > 1 \longrightarrow \sum_{n=0}^{\infty} u_n \longrightarrow$ divergent

Exercise

- [4.01] Examine the convergence of $\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + ... + z^n + ...$

$$S_n = 1 + z + z^2 + ... + z^{n-1} = \frac{1 - z^n}{1 - z}$$

$|z| < 1$ $\lim_{n \to \infty} S_n = \frac{1}{1 - z}$ convergent

$|z| \geq 1$ $|z^n| \geq 1$ divergent

Power series (幂级数)

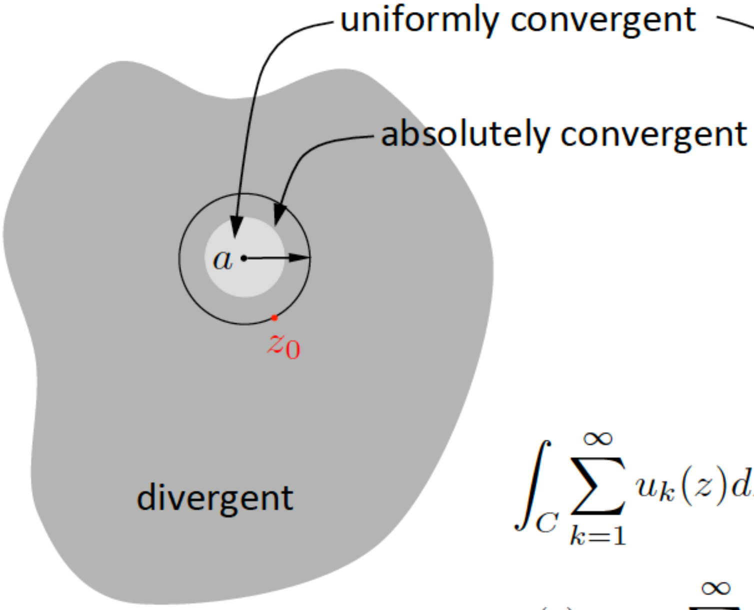
$$\sum_{n=0}^{\infty} c_n(z-a)^n = c_0 + c_1(z-a) + c_2(z-a)^2 + \dots$$

Abel theorem: If the series $\sum_{n=0}^{\infty} c_n(z-a)^n$ are convergent at $z = z_0$, then

the series are absolutely convergent in a disk region (with a radius of $|z_0 - a|$) surrounding z_0, and are uniformly convergent in the region $|z-a| \le r \ (r < |z_0 - a|)$.

Corollary: If $\sum_{n=0}^{\infty} c_n(z-a)^n$ are divergent at z_1 , then also divergent in $|z-a| > |z_1 - a|$.

Power series (幂级数)



$u_k(z) = c_k(z-a)^k$ is an analytic function defined in $\mathbb{C}$. The series are also uniformly convergent inside the radius of convergence.

$$\int_C \sum_{k=1}^{\infty} u_k(z)dz = \sum_{k=1}^{\infty} \int_C u_k(z)dz$$

$$f^{(p)}(z) = \sum_{k=1}^{\infty} u_k^{(p)}(z)$$

Exchange the order of operations

Taylor series (泰勒级数)

Theorem (Taylor expansion)

Assume $f(z)$ is analytic inside circle *C* (centered at point *a*) as well as along *C*, then for any point *z* inside *C*, $f(z)$ can be represented by a power series (expanded around *a*).

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n$$

where $a_n = \frac{1}{2\pi \mathrm{i}} \oint_C \frac{f(\zeta)}{(\zeta - a)^{n+1}} d\zeta = \frac{f^{(n)}(a)}{n!}$

Note: the contour integration is along the positive (anti-clockwise) direction of *C*.

$$e^z = 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^n}{n!} + \cdots = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \qquad |z| < \infty,$$

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n, \qquad |z| < 1.$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = \sum_{n=0}^{\infty} \frac{(-)^n}{(2n+1)!} z^{2n+1}, \qquad |z| < \infty.$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} = \sum_{n=0}^{\infty} \frac{(-)^n}{(2n)!} z^{2n}, \qquad |z| < \infty.$$

Compute Taylor Series

- 有理数化成部分分式

$$\frac{1}{1-3z+2z^2} = -\frac{1}{1-z} + \frac{2}{1-2z} = \sum_{n=0}^{\infty} \left(2^{n+1}-1\right) z^n, \quad |z| < \frac{1}{2}$$

- 级数相乘

$$f(z) = \frac{\mathrm{e}^z}{1-z}$$

- 求导法

$$\begin{aligned}\frac{1}{(1-z)^2} &= \frac{\mathrm{d}}{\mathrm{d}z}\frac{1}{1-z} = \frac{\mathrm{d}}{\mathrm{d}z}\sum_{n=0}^{\infty} z^n \\ &= \sum_{n=0}^{\infty}\frac{\mathrm{d}}{\mathrm{d}z}z^n = \sum_{n=1}^{\infty} n z^{n-1} \\ &= \sum (n+1) z^n \qquad |z| < 1\end{aligned}$$

- **Definition**

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n \qquad \text{where} \quad a_n = \frac{1}{2\pi \mathrm{i}} \oint_C \frac{f(\zeta)}{(\zeta - a)^{n+1}} d\zeta = \frac{f^{(n)}(a)}{n!}$$

$\exp\left\{\frac{1}{1-z}\right\}$, 在 $z=0$ 展开 (可只求前四项). $f'(z) = \frac{1}{(1-z)^2} e^{\frac{1}{1-z}}$, $f'(0) = e$ $\quad f''(z) = \left[\frac{2}{(1-z)^3} + \frac{1}{(1-z)^4}\right] e^{\frac{1}{1-z}}$, $f''(0) = 3e$

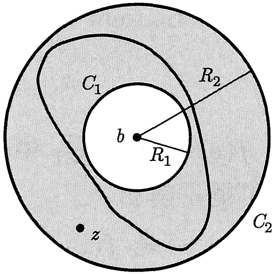
How about expansion at the singular point?

Laurent series (劳伦级数)

- Assume $f(z)$ is single-valued and analytic in a ring-like region $R_1 \le |z-b| \le R_2$, then for any z inside the ring, $f(z)$ can be expanded as a power series:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-b)^n, \quad R_1 < |z-b| < R_2$$

$$a_n = \frac{1}{2\pi \mathrm{i}} \oint_C \frac{f(\zeta)}{(\zeta-b)^{n+1}} d\zeta$$



Here *C* is an arbitrary closed curve inside the ring.

Theorem VII Laurent's theorem [equation (4.1)] (which we shall state without proof). Let C_1 and C_2 be two circles with center at z_0. Let $f(z)$ be analytic in the region R between the circles. Then $f(z)$ can be expanded in a series of the form

$$f(z) = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots + \frac{b_1}{z - z_0} + \frac{b_2}{(z - z_0)^2} + \cdots \tag{4.1}$$

convergent in R. Such a series is called a *Laurent series.* The "b" series in (4.1) is called the *principal part* of the Laurent series.

Exercise

Example Find the Laurent expansion of $\frac{1}{z(z-1)}$ when (i) $0<|z|<1$ and (ii) $|z|>1$

You should refer to the previous exercises.

convergent in $0<|z|<1$

(i) $0<|z|<1$

$$\frac{1}{z(z-1)} = -\frac{1}{z}\frac{1}{1-z} = -\frac{1}{z}\sum_{n=0}^{\infty} z^n = -\sum_{n=-1}^{\infty} z^n$$

When changing the region, the result changes.

And we can confirm that $\frac{1}{z(z-1)}$ has singular point at $z=0$.

(ii) $|z|>1$

$$\frac{1}{z(z-1)} = \frac{1}{z^2}\frac{1}{1-\frac{1}{z}} = \frac{1}{z^2}\sum_{n=0}^{\infty}\left(\frac{1}{z}\right)^n = \sum_{n=-2}^{-\infty} z^n$$

convergent in $|z|>1$

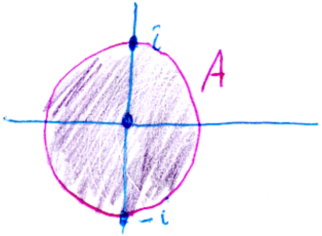
And we can confirm that $\frac{1}{z(z-1)}$ has singular point along $|z|=1$.

Example Compute the Laurent series for

$$f(z) = \frac{z+1}{z^3(z^2+1)}$$

on the region $A:\ 0 < |z| < 1$ centered at $z = 0$.

Solution: This function has isolated singularities at $z = 0, \pm i$. Therefore it is analytic on the region A.



$f(z)$ has singularities at $z = 0, \pm i$.

At $z = 0$ we have

$$f(z) = \frac{1}{z^3}(1+z)(1 - z^2 + z^4 - z^6 + \ldots).$$

Multiplying this out we get

$$f(z) = \frac{1}{z^3} + \frac{1}{z^2} - \frac{1}{z} - 1 + z + z^2 - z^3 - \ldots.$$

Zero Point

If $f(z)$ is analytic in a region around $z = a$, and is not always equal to zero. If $f(a) = 0$, then $z = a$ is called a zero point of $f(z)$.

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n, \ |z-a| < \rho$$

If $z = a$ is zero point, then $a_0 = a_1 = \ldots = a_{m-1} = 0, \quad a_m \neq 0$ $\quad (z-1)^{10}$

And we call $z = a$ the *m*-th order zero point of $f(z)$.

$$f(a) = f'(a) = \ldots = f^{(m-1)}(a) = 0, \quad f^{(m)}(a) \neq 0$$

If $z = a$ is the zero point of $f(z)$, and $f(z)$ is not always equal to zero for a small region around $z = a$, then $\exists \rho > 0$, such that $f(z)$ does not have zero points in $0 < |z-a| < \rho$.

Singular Point

$f(z)$ is analytic on $0 < |z - z_0| < r$ and has Laurent series

$$f(z) = \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} + \sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

Definition of poles. If only a finite number of the coefficients b_n are nonzero we say z_0 is a finite pole of f. In this case, if $b_k \neq 0$ and $b_n = 0$ for all $n > k$ then we say z_0 is a pole of order k.

- If z_0 is a pole of order 1 we say it is a simple pole of f.

本性奇点

- If an infinite number of the b_n are nonzero we say that z_0 is an essential singularity or a pole of infinite order of f.

可去奇点

- If all the b_n are 0, then z_0 is called a removable singularity. That is, if we define $f(z_0) = a_0$ then f is analytic on the disk $|z - z_0| < r$.

The coefficient b_1 of $1/(z - z_0)$ is called the *residue* of $f(z)$ at $z = z_0$.

Singular Points

可去奇点 由于在可去奇点处的级数展开不含负幂项，故级数不但在环域内收敛，而且在环域的中心，即可去奇点 $z=b$ 处也是收敛的．也就是说，这时的收敛区域其实是一个圆形区域，圆心位于可去奇点 $z=b$ 处，级数在收敛圆内内闭一致收敛，因而其和函数连续，

$$\lim_{z\to b} f(z) = \lim_{z\to b}\sum_{n=0}^{\infty} a_n(z-b)^n = a_0, \tag{5.33}$$

极点 函数在极点空心邻域内的 Laurent 展开有有限个负幂项，

$$\begin{aligned} f(z) &= a_{-m}(z-b)^{-m} + a_{-m+1}(z-b)^{-m+1} + \cdots + a_{-1}(z-b)^{-1} + a_0 + a_1(z-b) + \cdots \\ &= (z-b)^{-m}\left[a_{-m} + a_{-m+1}(z-b) + a_{-m+2}(z-b)^2 + \cdots\right] \\ &= (z-b)^{-m}\phi(z), \qquad 0<|z-b|<R, \end{aligned} \tag{5.35}$$

其中 m 是正整数，

$$\phi(z) = a_{-m} + a_{-m+1}(z-b) + a_{-m+2}(z-b)^2 + \cdots$$

在 $z=b$ 点的邻域内是解析的，而且 $\phi(b)=a_{-m}\neq 0$．b 点就称为 $f(z)$ 的 m **阶极点**．显然，只要 $|z-b|$ 足够小，$|f(z)|$ 可以大于任何正数，

$$\lim_{z\to b} f(z) = \infty. \tag{5.36}$$

本性奇点 函数在本性奇点邻域内的 Laurent 展开具有无穷多个负幂项.

如果 $z=b$ 是函数 $f(z)$ 的本性奇点，则当 $z\to b$ 时，$f(z)$ 的极限不存在. 更准确地说，$z\to b$ 的方式不同，$f(z)$ 可以逼近不同的数值. 例如，$z=0$ 是函数

$$\mathrm{e}^{1/z}=\sum_{n=0}^{\infty}\frac{1}{n!}\left(\frac{1}{z}\right)^{n},\quad 0<|z|<\infty$$

Example $\frac{\cos az}{z^2}, z=0$ two-order pole point

$\frac{\cos az-\cos bz}{z^2},\ a^2\neq b^2\ , z=0$ removable singularity

$$f(z)=z^{-2}\left[\left(1-\frac{1}{2!}a^2z^2+\frac{1}{4!}a^4z^4-+\cdots\right)-\left(1-\frac{1}{2!}b^2z^2+\frac{1}{4!}b^4z^4-+\cdots\right)\right]$$

$$=-\frac{1}{2!}\left(a^2-b^2\right)+\frac{1}{4!}\left(a^4-b^4\right)z^2-+\cdots$$

- Directly using Laurent series
- Using limit to check removable, pole, essential singularity

For pole, check its order by e.g., Laurent series

Chapter 6. Residual Theorem and Integration

Residual Theorem

integrals along the cuts cancel), or the integral along C is the sum of the integrals around the circles (all counterclockwise). But by (5.1), the integral around each circle is $2\pi i$ times the residue of $f(z)$ at the singular point inside. Thus we have the *residue theorem*:

(5.2) $$\oint_C f(z)\,dz = 2\pi i \cdot \text{sum of the residues of } f(z) \text{ inside } C,$$

where the integral around C is in the counterclockwise direction.

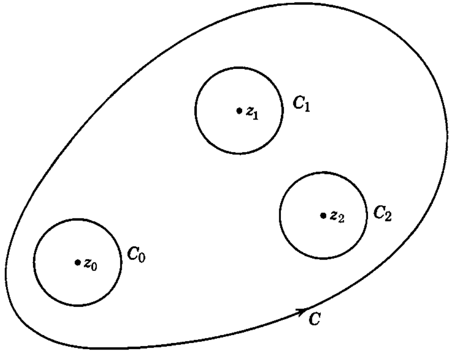


Figure 5.1

Summarize: Methods of Finding Residual

1. Laurent series: the coefficient b_1 (or a_{-1}) of the term $1/(z - z_0)$

2. Simple pole

$$R(z_0) = \lim_{z \to z_0} (z - z_0) f(z) \qquad \text{when } z_0 \text{ is a simple pole.}$$

$$R(z_0) = \frac{g(z_0)}{h'(z_0)} \qquad \text{if } \begin{cases} f(z) = g(z)/h(z), \text{ and} \\ g(z_0) = \text{finite const.} \neq 0, \text{ and} \\ h(z_0) = 0, \; h'(z_0) \neq 0. \end{cases}$$

3. Multiple poles: poles of order larger than $n > 1, m > n$

If the pole order is m, then

$$R(z_0) = a_{-1} = \lim_{z \to z_0} \left\{ \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)] \right\}$$

$$a_{-1} = \frac{1}{(m_0-1)!} \frac{d^{m_0-1}}{dz^{m_0-1}} (z - b)^{m_0} f(z)|_{z=b} \text{ with } m_0 \geq m$$

Example 4. Find the residue of $(\sin z)/(1-z^4)$ at $z=i$. By (6.2) we have

$$R(i)=\left.\frac{\sin z}{-4z^3}\right|_{z=i}=\frac{\sin i}{-4i^3}=\frac{e^{-1}-e}{(2i)(4i)}=\frac{1}{8}(e-e^{-1})=\frac{1}{4}\sinh 1.$$

Example 5. Find the residue of $f(z)=(z\sin z)/(z-\pi)^3$ at $z=\pi$.

We take $m=3$ to eliminate the denominator before differentiating; this is an allowed choice for m because the order of the pole of $f(z)$ at π is not greater than 3 since $z\sin z$ is finite at π. (The pole is actually of order 2, but we do not need this fact.) Then following the rule stated, we get

$$R(\pi)=\frac{1}{2!}\left.\frac{d^2}{dz^2}(z\sin z)\right|_{z=\pi}=\frac{1}{2}[-z\sin z+2\cos z]_{z=\pi}=-1.$$

$\left(\dfrac{z}{1-\cos z}\right)^2$, $z_0=0$;

$$1-\cos z=1-\left(1-\frac{1}{2!}z^2+\frac{1}{4!}z^4-\frac{1}{6!}z^6+-\cdots\right)=\frac{1}{2}z^2\left(1-\frac{1}{12}z^2+\frac{1}{360}z^4-+\cdots\right)$$

Integration

- Complex line integrate based on $\gamma(t)$
- Cauchy's Theorem
- Cauchy's integral formular
- Residual theorem (general)

Theorem VI Cauchy's integral formula (which we shall prove). If $f(z)$ is analytic on and inside a simple closed curve C, the value of $f(z)$ at a point $z = a$ inside C is given by the following contour integral along C:

$$f(a) = \frac{1}{2\pi i} \oint \frac{f(z)}{z-a} dz.$$

$$\oint_C f(z)\, dz = 2\pi i \cdot \text{sum of the residues of } f(z) \text{ inside } C, \tag{5.2}$$

where the integral around C is in the counterclockwise direction.

[5.02] Find the value of $\oint_C z^n dz$, where *n* is an integer, *C* is a simply closed curve in $\mathbb{C}$.

- Based on the Cauchy integral formula, we can deduce a very important conclusion:

If $f(z)$ is analytic in a bounded region $\overline{G}$, then any higher-order derivative of $f(z)$ exists, which is given by

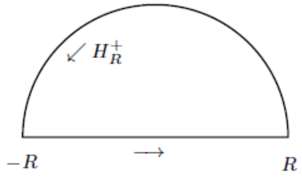
$$f^{(n)}(z) = \frac{n!}{2\pi \mathrm{i}} \oint_C \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta$$

$$R(z_0) = a_{-1} = \lim_{z \to z_0} \left\{ \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)] \right\}$$

Integration

1. 三角函数$\boldsymbol{\theta(0, 2\pi)}$的积分化为积分
2. $(-\infty,+\infty)$的积分化为回路积分
   - 分母higher than 2 degrees
   - higher than 1 degree
   - 分母有zeros
3. Singular points along the path
4. Integration with a multi-valued function

- Jordan's Theorem



Jordan's Lemma deals with the problem of how a contour integral behaves on the semi-circular arc H_R^+ of a closed contour C.

Lemma 1 *(Jordan) If the only singularities of $F(z)$ are poles, then*

$$\lim_{R\to\infty} \int_{H_R} e^{imz} F(z)\, dz = 0 \tag{1}$$

provided that $m > 0$ and $|F(z)| \to 0$ as $R \to \infty$.

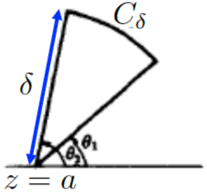
a) When $m > 0$ forms of $F(z)$ such as $F(z) = \frac{1}{z}$, $F(z) = \frac{1}{z+a}$ or rational functions of z such as $F(z) = \frac{z^p \dots}{z^q + \dots}$ (for $0 \le p < q$ and p and q integers) will all work as these all have simple poles and $|F(z)| \to 0$ as $R \to \infty$.

b) If, however, $m = 0$ then a modification is needed: e.g. if $F(z) = \frac{1}{z}$ then $|F(z)| \to 0$ but the $R|F(z)| = 1$. We need to alter the restriction on the integers p and q to $0 \le p < q - 1$ which excludes cases like $F(z) = \frac{1}{z}$, $F(z) = \frac{1}{z+a}$.

Small arc lemma (小圆弧引理)

If $f(z)$ is continuous in a small region around $z=a$, and satisfies the following relation: when $\theta_1 \le \arg(z-a) \le \theta_2$, $|z-a| \to 0$, $(z-a)f(z)$ uniformly approaches k. Then we have

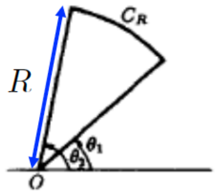
$$\lim_{\delta \to 0} \int_{C_\delta} f(z)dz = \mathrm{i}k(\theta_2 - \theta_1)$$



Great arc lemma (大圆弧引理)

If $f(z)$ is continuous in a region around $z=\infty$, and satisfies the following relation: when $\theta_1 \le \arg(z) \le \theta_2$, $z \to \infty$, $zf(z)$ uniformly approaches K. Then we have

$$\lim_{R \to \infty} \int_{C_R} f(z)dz = \mathrm{i}K(\theta_2 - \theta_1)$$



1. 三角函数$\theta(0, 2\pi)$的积分化为积分

Example 1. Find $I = \displaystyle\int_0^{2\pi} \frac{d\theta}{5 + 4\cos\theta}$.

This method can be used to evaluate the integral of any rational function of sinθ and cosθ between 0 and 2π, provided the denominator is never zero for any value of θ.

$$I = \int_0^{2\pi} R(\sin\theta,\ \cos\theta) d\theta$$

$R(,)$ is continuous in $[0, 2\pi]$

$$z = e^{\mathrm{i}\theta} \qquad \frac{1}{z} = e^{-\mathrm{i}\theta}$$

Variable substitution

$$I = \oint_{|z|=1} R\left(\frac{z^2-1}{2\mathrm{i}z}, \frac{z^2+1}{2z}\right)\frac{dz}{\mathrm{i}z}$$

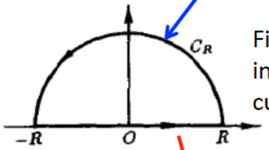
$R(,)$ does not have any singular point along the unit circle $|z| = 1$; it may have singular points inside the circle

2. (−∞, ∞)的积分化为(半平面)回路积分

Example 2. Evaluate $I = \int_{-\infty}^{\infty} \frac{dx}{1+x^2}$.

$$I = \int_{-\infty}^{\infty} f(x)dx$$

Judge the behavior along the great arc



Find the residue(s) inside the closed curve

$$I = \lim_{R_1\to\infty, R_2\to\infty} \int_{-R_1}^{R_2} f(x)dx$$

$$\text{v.p.} \int_{-\infty}^{\infty} f(x)dx = I = \lim_{R\to\infty} \int_{-R}^{R} f(x)dx$$

积分主值

The basic idea is to expand the region of integration.

(1) $f(z)$ 在上半平面除有限个孤立奇点外处处解析，在实轴上没有奇点；

(2) 在 $0 \leqslant \arg z \leqslant \pi$ 范围内，当 $|z| \to \infty$ 时，$zf(z)$ 一致地趋于 0，即 $\forall \varepsilon > 0$，$\exists M(\varepsilon) > 0$，使当 $|z| \geqslant M$，并且 $0 \leqslant \arg z \leqslant \pi$ 时，$|zf(z)| < \varepsilon$.

See Great arc lemma

3. 含三角函数的(−∞,+∞)积分

Example 3. Evaluate $I = \int_0^\infty \frac{\cos x\, dx}{1+x^2}$.

$$I = \int_{-\infty}^{\infty} f(x)\cos px dx,\ \ I = \int_{-\infty}^{\infty} f(x)\sin px dx \qquad (p>0)$$

We do not take the substitution $f(z)\cos pz$ or $f(z)\sin pz$, because $z=\infty$ is an essential singular point.

Z=1/t, Check from the number of b_n for $\sin(\frac{1}{pt})$

Instead, we take $f(z)e^{ipz}$

$$\oint_C f(z)e^{ipz}dz = \int_{-R}^{R} f(x)(\cos px + i\sin px)dx + \int_{C_R} f(z)e^{ipz}dz$$

4. Singular points along the path

Example 4. Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{\sin x}{x}\, dx.$ $\displaystyle\int \frac{e^{iz}}{z}\, dz.$

We have to get around those singular points, and apply the limit of improper integral.

$$\int_a^b f(x)dx = \lim_{\delta_1 \to 0} \int_a^{c-\delta_1} f(x)dx + \lim_{\delta_2 \to 0} \int_{c+\delta_2}^{b} f(x)dx$$

$$\text{v.p.} \int_a^b f(x)dx = \lim_{\delta \to 0} \left[\int_a^{c-\delta} f(x)dx + \int_{c+\delta}^{b} f(x)dx \right]$$

Usually, the singular point is of 1^{st}-order pole, then we can apply the small/great arc lemma.

$$\lim_{|z-a| \to 0} (z-a)f(z) \to k \quad \text{or} \quad \lim_{z \to \infty} zf(z) \to K$$

If the order is much higher, then we may not apply the small/great arc lemma.